

Exercise n. 3

Write a numerical code to solve the **2D incompressible Navier-Stokes equations**, incorporating specific **initial and boundary conditions**, within a rectangular domain:

$$\begin{cases} \frac{\partial u_i}{\partial t} = -\frac{\partial}{\partial x_j}(u_i u_j) - \frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j}, \\ \frac{\partial u_i}{\partial x_i} = 0. \end{cases}$$

where u and v are the horizontal and vertical velocity components, respectively, ν the kinematic viscosity and ρ the density.

The problem should be solved using a **three-stage Runge-Kutta method** for time integration and **second-order finite-difference stencils for spatial discretization** of the differential operators. You may either start from scratch or complete the provided code template. Once the code is ready, carry out the tasks described in the following.

1) Validation of the diffusive terms - Poiseuille flow

Apply the following boundary conditions:

- wall on $y = 0$ ($u = 0$, $v = 0$ and $\partial p / \partial y = 0$);
- wall on $y = L_y$ ($u = 0$, $v = 0$ and $\partial p / \partial y = 0$);
- inflow on $x = 0$ ($\partial p / \partial x = 0$ and $u(y) = 1$)
- outflow on $x = L_x$ (pressure $p = 0$ e $\partial u / \partial x = 0$ $\partial v / \partial x = 0$).

Apply zero initial conditions for both velocity and pressure fields.

Consider a rectangular computational domain of dimensions $L_x \times L_y = 10 \times 1$. Discretize the domain using at least $N_x \times N_y = 100 \times 40$ grid nodes. Set the density to 1 and set the Reynolds number to $Re = U_b h / \nu = 500$, with $U_b = 1$ the bulk velocity (volumetric average of the velocity field over the entire computational domain) and $h = L_y = 1$ the channel height.

Compute the solution over the time interval $t \in [0, 20]$, using a time step $\Delta t = 10^{-2}$ to ensure numerical stability and accuracy. It is recommended to store the solution fields at equispaced time intervals. After completing the simulation, perform the following sub-tasks:

- Using Paraview (advised), MatLab or Python **visualize and provide the view of the velocity field** (u and v) at selected time steps (e.g., $t = 5, 10, 15, 20$).

- **Plot the time evolution of the horizontal velocity component sampled along a vertical line** located at the at $x = 9.5$. Generate plots at three or four selected time instants (e.g., $t = 5, 10, 15, 20$). Verify that, by the end of the simulation, the velocity profile approaches a parabolic distribution according to the Poiseuille flow theory, $u(y) = 6.0y(h - y)$.
-

2) Validation - Taylor-Green vortex benchmark

The Taylor–Green vortex is a classical benchmark for incompressible Navier-Stokes solvers. It consists of a smooth, periodic velocity field made of counter-rotating vortices arranged in a regular pattern. The initial condition is divergence-free by construction, making it ideal for testing projection methods. As the flow evolves, viscosity causes the vortices to gradually decay, leading to a monotonic decrease of kinetic energy. Because the exact initial structure is well defined and highly symmetric, this test case is commonly used to assess numerical accuracy, conservation properties, and the correct enforcement of incompressibility and periodic boundary conditions.

Consider a rectangular computational domain of dimensions $L_x \times L_y = 2\pi \times 2\pi$. Discretize the domain using at least $N_x \times N_y = 64 \times 64$ grid nodes.

Apply the following boundary conditions:

- Periodic on x ;
- Periodic on y ;

and the following initial conditions:

$$\begin{aligned} u(x, y) &= U_0 \sin(kx) \cos(ky), \\ v(x, y) &= -U_0 \cos(kx) \sin(ky), \\ p(x, y) &= \frac{\rho U_0^2}{4} [\cos(2kx) + \cos(2ky)], \end{aligned}$$

where u and v are the horizontal and vertical velocity components, respectively and p the pressure. U_0 is an initial velocity magnitude that can be set to $U_0 = 1$, whereas the wavenumber k can be set to $k = 1$, too. Set the viscosity to $\nu = 10^{-2}$ and consider unitary density, $\rho = 1$. Compute the solution over the time interval $t \in [0, 75]$, using a time step $\Delta t = 10^{-2}$ to ensure numerical stability and accuracy.

Perform the following tasks:

- Using Paraview (advised), MatLab or Python visualize and provide the view of the velocity field (u and v) at selected time steps (e.g., $t = 25, 50, 75$);
- Compare the obtained numerical solution with the analytical one. To this purpose, you can provide, at different time steps, the plot of the velocity and pressure fields along a horizontal or

vertical line on the half of the computational domain. The 2D exact, reference solution is given by:

$$\begin{aligned}u(x, y, t) &= U_0 \sin(kx) \cos(ky) e^{-2\nu k^2 t}, \\v(x, y, t) &= -U_0 \cos(kx) \sin(ky) e^{-2\nu k^2 t}, \\p(x, y, t) &= \frac{\rho U_0^2}{4} [\cos(2kx) + \cos(2ky)] e^{-4\nu k^2 t}.\end{aligned}$$

3) OPTIONAL

Choose an arbitrary set of initial and boundary conditions and perform a numerical simulation of the selected test case.

Provide a set of representative snapshots of the solution, including velocity and pressure fields at different times. In addition, include some quantitative analysis of the flow evolution, such as (but not mandatory):

- o maximum velocity magnitude and its evolution in time;
- o velocity and pressure profiles over lines;
- o divergence of the velocity field (to assess incompressibility);
- o qualitative observations on flow structures (e.g. vortices, shear layers, instabilities).

Briefly discuss the physical behavior of the solution and comment on the numerical performance of the method (stability, accuracy, and sensitivity to grid resolution or time step).

Tips (suggested test cases):

- o **Kelvin–Helmholtz instability.** Consider a shear layer with two streams moving at different velocities. A small perturbation added to the interface will trigger the growth of vortices due to the velocity discontinuity. This test is useful to assess the ability of the solver to capture instabilities, vortex roll-up, and nonlinear flow evolution.
- o **Top-hat jet.** Initialize a jet with a uniform velocity profile (sharp or smoothed transition) embedded in a quiescent fluid. This configuration generates shear layers at the jet boundaries, which may develop into vortical structures. It is a good test for convection, diffusion, and numerical stability in the presence of strong gradients.